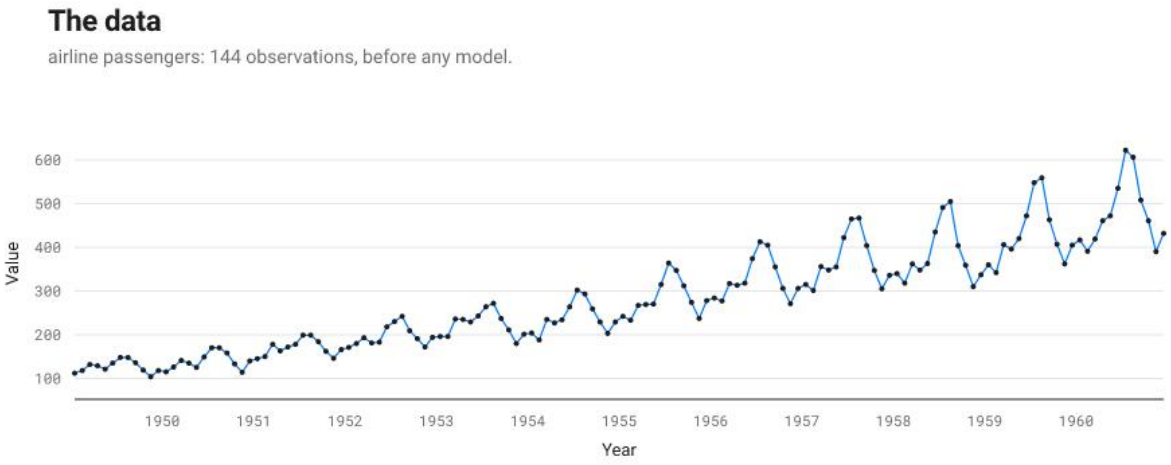
An automatic report for the dataset: airline passengers

Some Math empowered by some AI guided by some researchers

09 November 2026

Produced by **`autostat`***, a Python 3 port of the Automatic Bayesian Covariance Discovery system of Lloyd et al. (AAAI 2014).*



The data

Executive summary

The structure search selected `Lin x Per + (Lin + RQ) x (Lin + Lin x Per)` (Bayesian information criterion 1112.4; negative log likelihood 521.4 on 144 points). It decomposes into 5 additive components, listed by how much each reduces extrapolation error:

1. A quadratic polynomial.
2. This simple AI is not capable of describing the component Lin x RQ.
3. A periodic function with a period of 1.0 years but with quadraticly varying amplitude.
4. This simple AI is not capable of describing the component Lin x Per x RQ.
5. A periodic function with a period of 4.2 years but with linearly increasing amplitude.

#	R^2 cumulative (%)	Residual variance explained (%)	CV MAE	MAE reduction (%)
data			280	
1	81.6	81.6	106	62.1
2	86.6	27.1	32.3	69.5
3	97.1	78.6	16.5	48.9
4	99.6	87.3	8.57	48.1
5	100.0	100.0	5.84	31.9

MAE: mean absolute error over ten contiguous held-out blocks; CV: cross-validated.

The formula found

The structure search is a symbolic regression: it searches a grammar of expressions over kernels and keeps the one with the lowest Bayesian information criterion. This is the expression it kept, with every fitted parameter in place; t is in years.

$$y(t) = f_1(t) + f_2(t) + f_3(t) + f_4(t) + f_5(t) + \varepsilon(t)$$

Each component is a Gaussian process with its own covariance:

$$f_1(t) \sim \mathrm{GP}\big(0,\ \ 0.7107\,(t-1943)(t'-1943)\cdot(t-1940)(t'-1940)\big)$$

$$f_2(t) \sim \mathrm{GP}\left(0,\ \ 56.66\,(t-1940)(t'-1940)\cdot\left(1+\frac{(t-t')^2}{2\cdot 1.654\times 10^{-4}\cdot 0.001875^2}\right)^{-1.654\times 10^{-4}}\right)$$

$$f_3(t) \sim \mathrm{GP}\left(0,\ \ 0.04947\,(t-1943)(t'-1943)\cdot(t-1944)(t'-1944)\cdot\frac{\exp\Big(2.195\cos\frac{2\pi(t-t')}{1.002}\Big)-I_0(2.195)}{e^{2.195}-I_0(2.195)}\right)$$

$$f_4(t) \sim \mathrm{GP}\left(0,\ \ 3.944\,(t-1944)(t'-1944)\cdot\frac{\exp\Big(2.195\cos\frac{2\pi(t-t')}{1.002}\Big)-I_0(2.195)}{e^{2.195}-I_0(2.195)}\cdot\left(1+\frac{(t-t')^2}{2\cdot 1.654\times 10^{-4}\cdot 0.001875^2}\right)^{-1.654\times 10^{-4}}\right)$$

$$f_5(t) \sim \mathrm{GP}\left(0,\ \ 0.4332\,(t-1943)(t'-1943)\cdot\frac{\exp\Big(1.132\cos\frac{2\pi(t-t')}{4.18}\Big)-I_0(1.132)}{e^{1.132}-I_0(1.132)}\right)$$

$$\varepsilon(t) \sim \mathrm{N}\Big(0,\ \ 1.091\times 10^{-10}\Big)$$

These components have a closed form, read from their posterior mean:

$$f_1(t) = 48.88 + 13.61\,(t-1949) + 0.9081\,(t-1949)^2$$

The other components are smooth, quasi-periodic or noise without a short formula: a draw from such a process has no finite closed form; its covariance above is the formula.

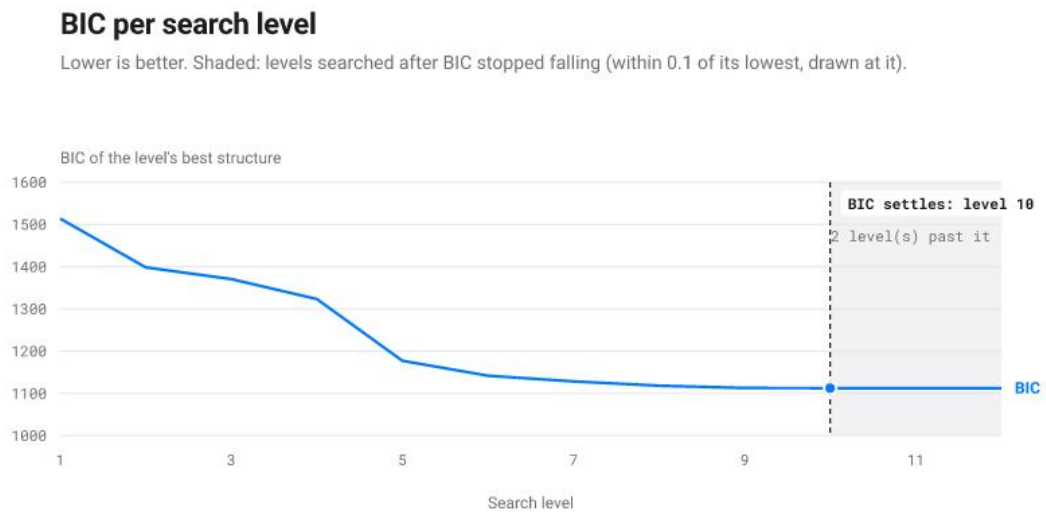
Closed form of each component:

#	component	closed form
1	A quadratic polynomial	exact polynomial of degree 2
2	This simple AI is not capable of describing the component Lin x RQ	none: the covariance is the formula
3	A periodic function with a period of 1.0 years but with quadraticly varying amplitude	none: the covariance is the formula
4	This simple AI is not capable of describing the component Lin x Per x RQ	none: the covariance is the formula
5	A periodic function with a period of 4.2 years but with linearly increasing amplitude	none: the covariance is the formula

Search path and stopping criterion

The 2014 configuration searches a fixed number of levels and keeps the lowest BIC; it has no stopping rule. The table follows the winner of each level. A change of structure adds or removes terms; a level that keeps the structure only refits its parameters. BIC differences read on the scale of Kass and Raftery (1995): under 2 barely worth mentioning, 2 to 6 positive, 6 to 10 strong, over 10 very strong.

level	BIC	change	structure	components	closed	params
1	1513.7		RQ	1	0	3
2	1398.4	-115.3	Lin + RQ	2	1	5
3	1370.4	-28.0	Lin x RQ + RQ	2	0	7
4	1323.2	-47.2	(Lin + Per) x RQ + RQ	3	0	10
5	1177.0	-146.1	(Lin + Per x RQ) x RQ + RQ	3	0	12
6	1141.8	-35.3	(Lin + Per x RQ) x (Lin + RQ) + RQ	5	1	14
7	1128.3	-13.5	(Lin + Lin x Per) x (Lin + RQ) + RQ	5	1	13
8	1118.0	-10.3	(Lin + Lin x Per) x (Lin + RQ) + C	5	2	11
9	1112.7	-5.3	(Lin + Lin x Per) x (Lin + RQ) + Per	5	2	13
10	1112.4	-0.2	(Lin + Lin x Per) x (Lin + RQ) + Lin x Per	5	1	14
11	1112.4	-0.0	(Lin + Lin x Per) x (Lin + RQ) + Lin x Per	5	1	14
12 (kept)	1112.4	-0.0	(Lin + Lin x Per) x (Lin + RQ) + Lin x Per	5	1	14

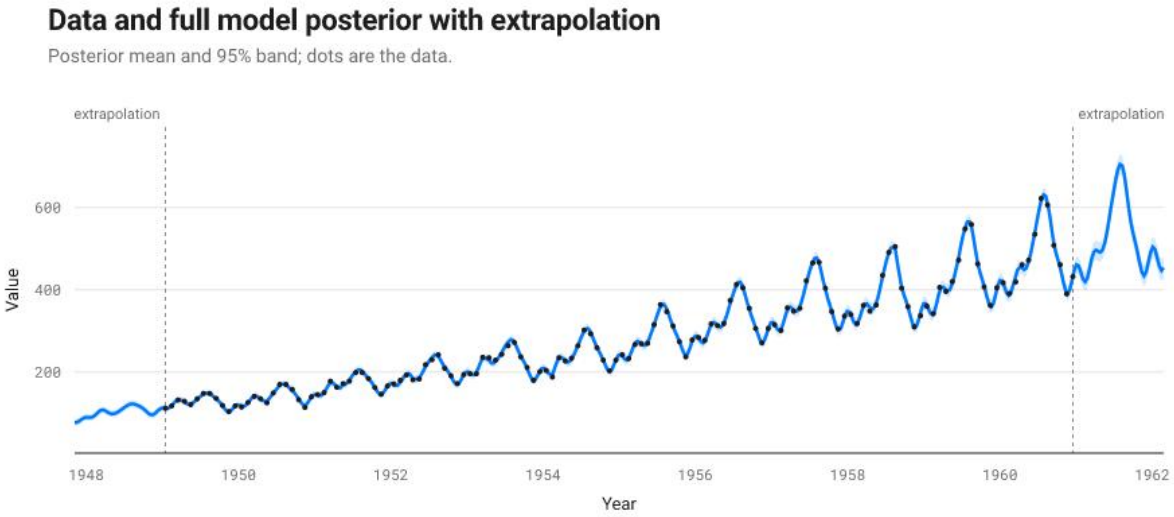


BIC per search level

Posterior weight of each structure met, exp(-BIC/2) normalised under a flat prior, after the Bayesian machine scientist (Guimerà et al., 2020): where the search keeps one structure, the weights say how much the data prefer it to the others. The 7 structure(s) under 0.001 are left out.

structure	lowest BIC	weight
(Lin + Lin x Per) x (Lin + RQ) + Lin x Per	1112.4	0.513
(Lin + Lin x Per) x (Lin + RQ) + Per	1112.7	0.455
(Lin + Lin x Per) x (Lin + RQ) + C	1118.0	0.032

- The search ran 12 levels (stopping after 2 level(s) without a BIC fall above 0.1, at most 20) and keeps the model of level 12, (Lin + Lin x Per) x (Lin + RQ) + Lin x Per, BIC 1112.4.
- That structure was first reached at level 10; the levels after it only refitted its parameters, lowering BIC by 0.0 in all.
- The best different structure, (Lin + Lin x Per) x (Lin + RQ) + Per (level 9), is 0.2 above it: barely worth mentioning evidence for the kept one on the Kass and Raftery scale.
- Weighed by exp(-BIC/2) under a flat prior, as the Bayesian machine scientist weighs expressions, the structures met give the kept one a posterior weight of 0.513 and the next, (Lin + Lin x Per) x (Lin + RQ) + Per, 0.455: the data barely tell them apart.
- BIC stopped falling at level 10. The 2 level(s) searched past it kept its structure and lowered BIC by 0.00 in all, less than the tolerance of 0.1: going on buys nothing the data pay for.
- A rule stopping after 1 level(s) without a fall above 0.1 would have ended the search at level 11 and kept the same structure, less refitted.
- A rule stopping after 2 level(s) without a fall above 0.1 would have ended the search at level 12 and kept the same model.
- The kept expression has 14 parameters for 144 points: its BIC is a fit term 2 NLL = 1042.9 plus a complexity term p ln n = 69.6, each extra parameter costing 4.97; it splits into 5 components, 1 of them with a closed form.



Data and full posterior

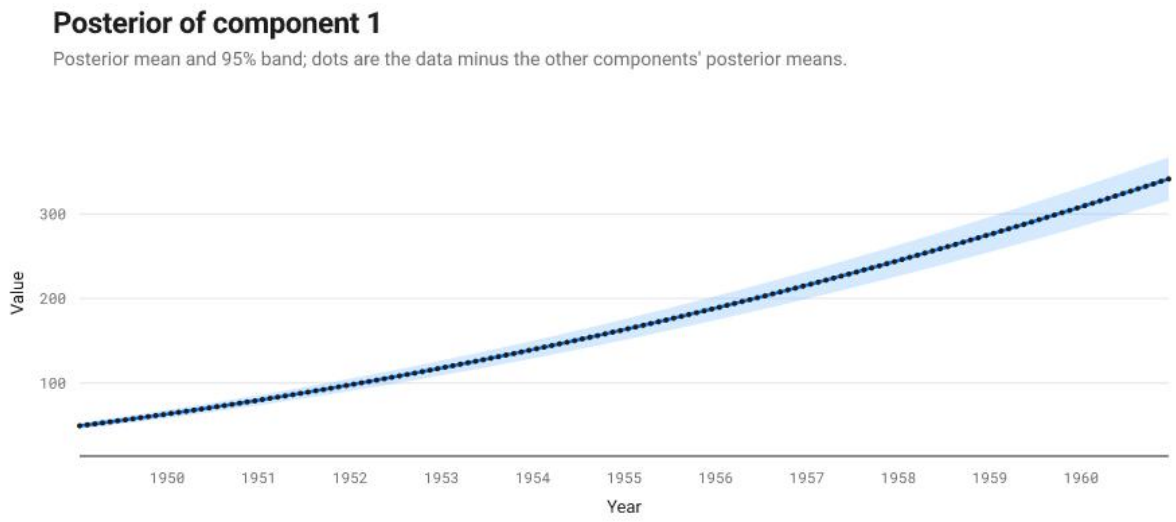
Detailed discussion of the components

Component 1: A quadratic polynomial

This component is a quadratic polynomial.

This component explains 81.6% of the data's variance on its own; with the components above it, 81.6%. Adding it changes the cross-validated MAE by -62.1%.

Extrapolation. This component is assumed to continue as a quadratic polynomial.



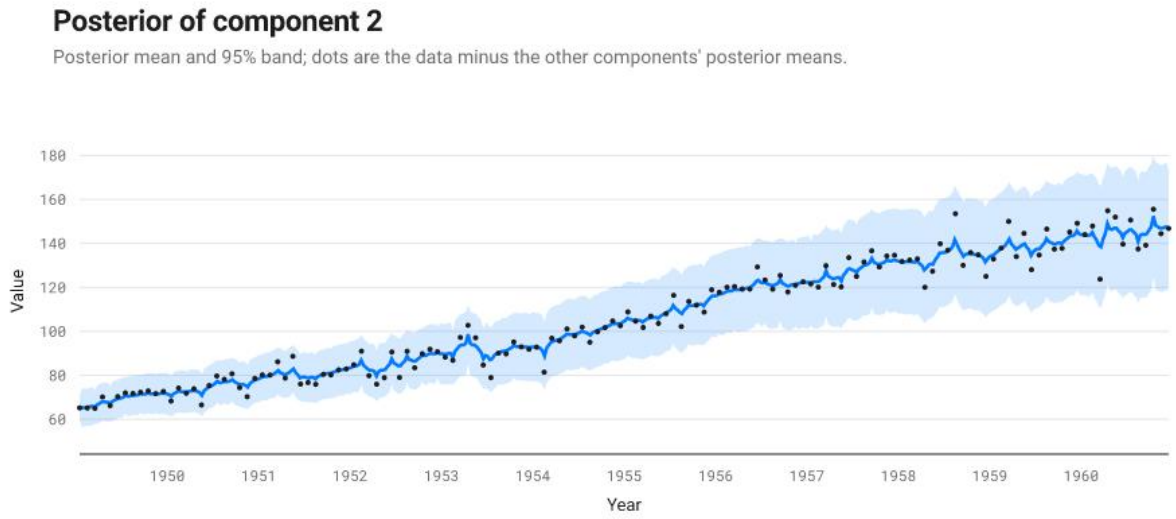
Component 1

Component 2: This simple AI is not capable of describing the component Lin x RQ

This simple AI is not capable of describing the component Lin x RQ.

This component explains 35.3% of the data's variance on its own; with the components above it, 86.6%. Adding it changes the cross-validated MAE by -69.5%.

Extrapolation. This simple AI is not capable of describing the component Lin x RQ.



Component 2

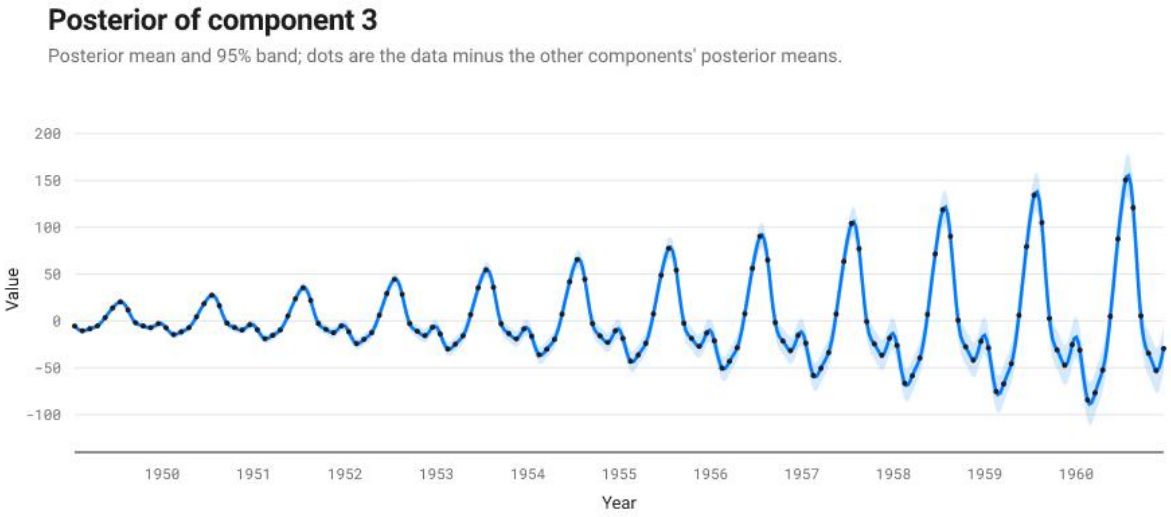
Component 3: A periodic function with a period of 1.0 years but with quadraticly varying amplitude

This component is periodic with a period of 1.0 years but with varying amplitude. The amplitude of the function varies quadraticly. The shape of this function within each period has a typical

lengthscale of 5.6 weeks.

This component explains 11.9% of the data's variance on its own; with the components above it, 97.1%. Adding it changes the cross-validated MAE by -48.9%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years but with varying amplitude. The amplitude of the function is assumed to continue to vary quadraticly.



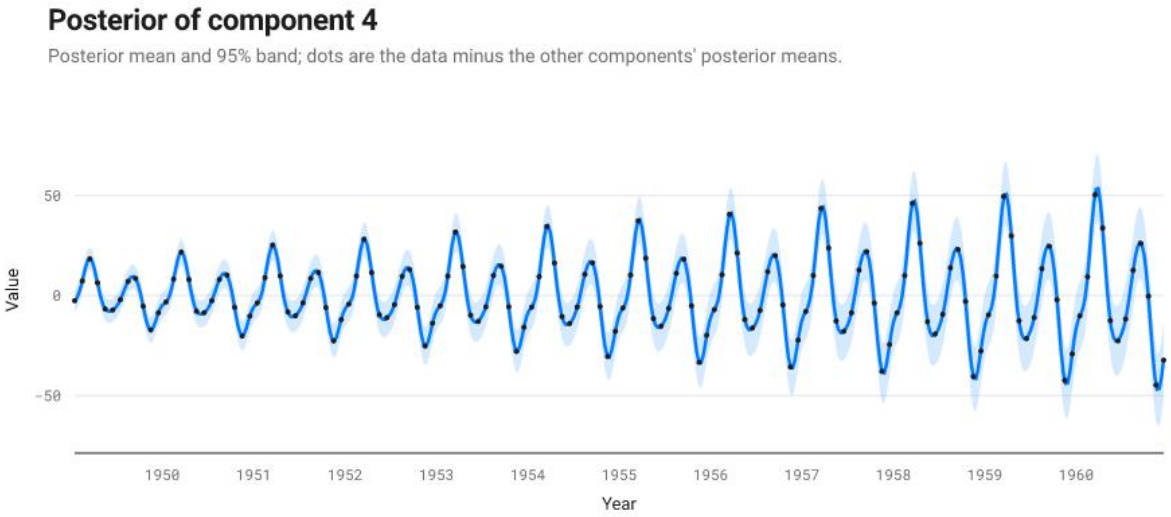
Component 3

Component 4: This simple AI is not capable of describing the component Lin x Per x RQ

This simple AI is not capable of describing the component Lin x Per x RQ.

This component explains -0.8% of the data's variance on its own; with the components above it, 99.6%. Adding it changes the cross-validated MAE by -48.1%.

Extrapolation. This simple AI is not capable of describing the component Lin x Per x RQ.



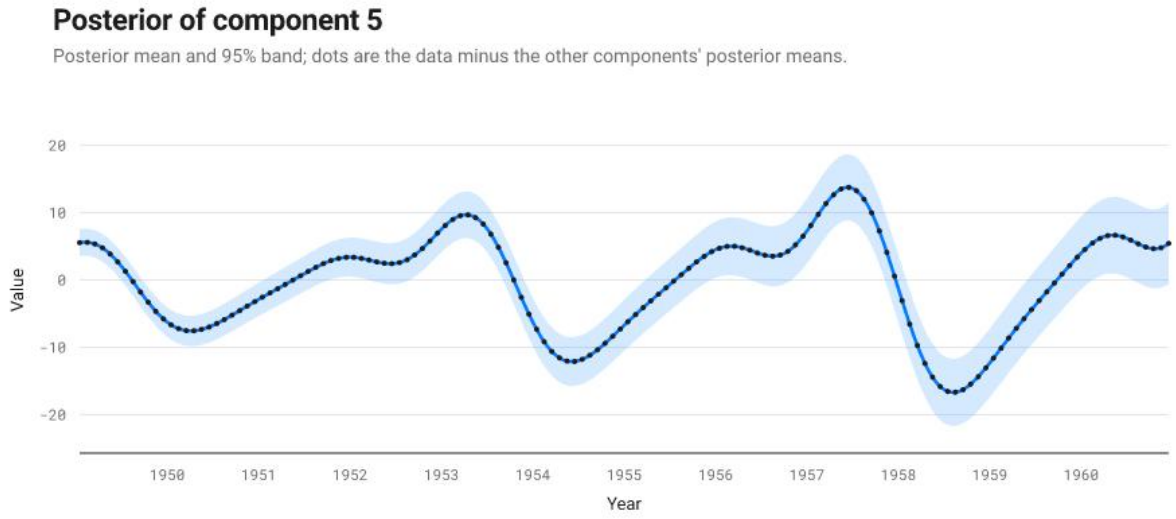
Component 4

Component 5: A periodic function with a period of 4.2 years but with linearly increasing amplitude

This component is periodic with a period of 4.2 years but with varying amplitude. The amplitude of the function increases linearly. The shape of this function within each period has a typical lengthscale of 7.5 months.

This component explains 0.6% of the data's variance on its own; with the components above it, 100.0%. Adding it changes the cross-validated MAE by -31.9%.

Extrapolation. This component is assumed to continue periodically with a period of 4.2 years but with varying amplitude. The amplitude of the function is assumed to continue to increase linearly.



Component 5

Model checking

Each statistic compares draws of a component's posterior with draws of its prior. A small p-value (bold below 0.05) points at a property of the data that the component's prior does not reproduce.

#	QQ max	QQ min	ACF min	ACF min lag	Pgram max	Pgram peak	MMD
1	0.284	0.284	0.982	0.988	0.564	0.974	**0.000**
2	0.324	0.344	0.496	0.774	0.672	0.968	**0.000**
3	0.564	0.272	0.784	0.824	0.522	0.638	0.087
4	0.958	0.580	0.770	0.554	0.904	0.412	**0.016**
5	0.424	0.848	0.824	0.792	0.964	0.682	**0.003**

QQ: quantile-quantile gaps. ACF: autocorrelation function. Pgram: periodogram. MMD: maximum mean discrepancy two-sample test.